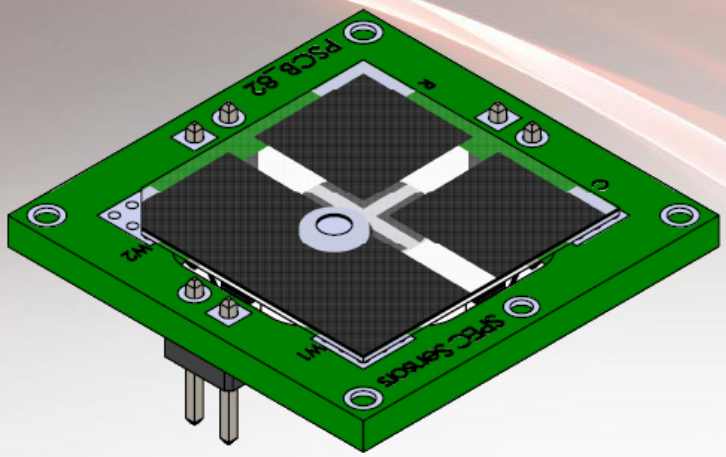
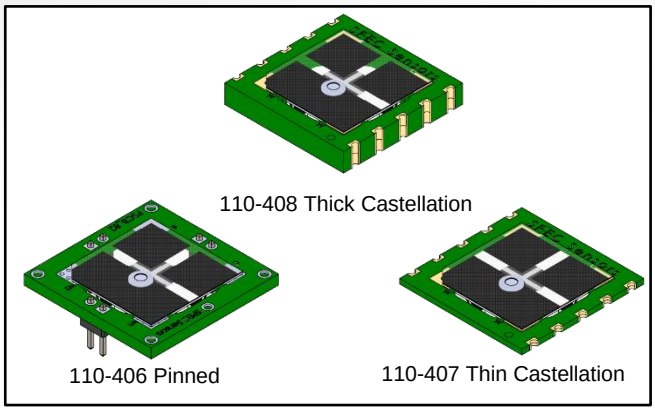
INTERLINK ELECTRONICS | ie SENSORS

PRODUCT BRIEF

Ozone (O3) Sensor

110-4xx





Interlink Electronics Screen Printed Electrochemical sensor (SPEC Sensor™) revolutionizes the current state of the art, enabling new applications in consumer, medical and industrial safety. Interlink Electronics offer the performance of the best quality electrochemical sensors at a fraction of the price. The 110-4xx family Ozone sensors are small and low-profile, facilitating easy integration into wireless, portable, and other IoT solutions. These sensors are ideal for health, food, environmental, industrial, and residential monitoring, because of their high performance, low cost, and small size. IE'S Ozone Sensor are available in three packages (110-406, 110-407, 110-408).

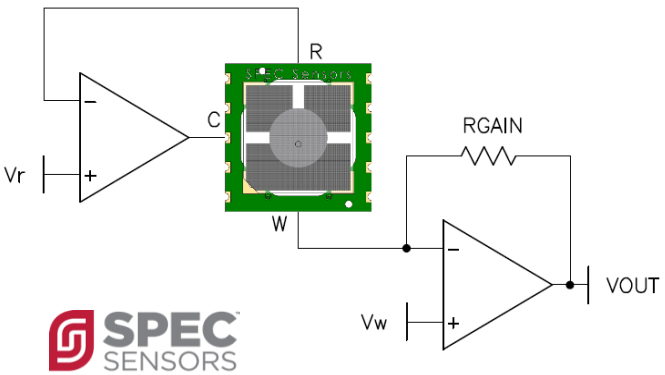
FEATURES

- Small Size & Low Profile (20 x 20 x 3 mm)
- Improved stability and low ppb sensitivity
- Accurate & Linear Response
- Long Life (10 +years)
- Fast Response
- 100% Factory Tested
- Ultra-Low Power < 50 uW max
- ROHS Compliant

APPLICATIONS

- Outdoor Air Quality
- Indoor Air Quality
- Food Quality
- Health/ Sanitizing
- Weather Stations
- Smart Homes
- Air Purification Control
- Industrial Safety

The diagram below shows the basic measuring circuit for the 110-4xx. The sensor generates an electric current which is converted into output voltage (Vout) using a linear circuit like the one shown.
Notes: Vbias = Vw-Vr



SPEC SENSORS

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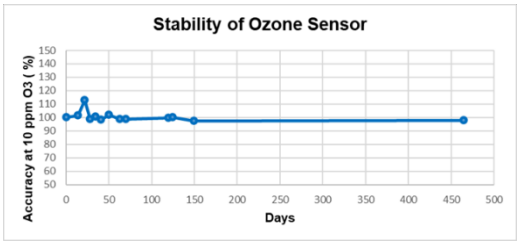
SPECIFICATIONS

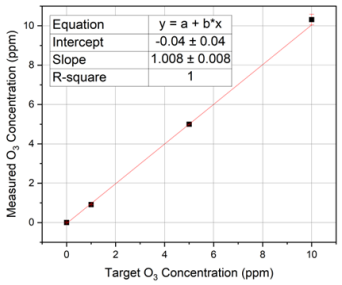
Parameter	Typical Value	Unit	Notes
Measurement Range	0 to 10	ppm	1,2,3
Lower Detection Limit	0.02	ppm	1,2
Resolution	0.01	ppm	1,2
Accuracy	± 2	%	1
Response Time – T (90)	< 180	seconds	1
Sensitivity Range	-50 ± 25	nA/ppm	1
Expected Operating Life	10	years	1
Operating Temperature Range	-30 to 55	°C	3
Operating Humidity Range	10 to 95	%	3,4
Operating Pressure Range	0.8 to 1.2	atm	3
Operating Bias Range	0 to -25	mV	
Power Consumption	10 to 50	uW	2

1. *At standard conditions (25 °C, 50% RH, 1 atm), unless otherwise indicated.*
2. *Depends on circuit design.*
3. *Recommended operating range. The sensor may be damaged, and warranty voided if operated outside the specified range. For the best optimum accuracy operate sensor at -20 to 40 °C and 15 to 90% RH.*
4. *Non-condensing.*

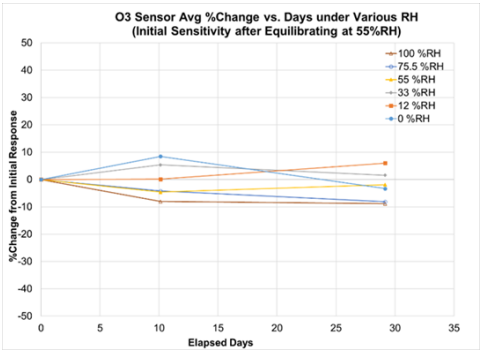
KEY CHARACTERISTICS



Long Term Stability



Linearity



Humidity Effect

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